

Quiz 2 solutions

Q1

2 Points

Adding regularization can help performance on the (held-out) test set because it helps reduce *variance* of the estimated function

☒ True

☐ False

Not much to say here, that's the goal.

Q2

2 Points

Consider ridge regression with parameter λ on the penalty term. When $\lambda > 0$, the model will usually have lower *training* error than the model with $\lambda = 0$.

☐ True

☒ False

We know that, when $\lambda = 0$, the ridge regression objective is exactly minimizing the training error. Adding any $\lambda > 0$ compromises between that goal and having a smaller β , which makes the training error worse.

Q3

2 Points

Which of the following advantages does ridge regression have over linear regression?
Select any that apply.

☐ Exact zeros in the solution for $\hat{\beta}$

☐ Ability to better approximate non-linear functions

☒ Reduced variance

☐ Lower bias

☐ Ability to automatically detect interactions

The ridge penalty does not introduce sparsity, since we've seen that the ridge solutions are always dense. The solution is still a linear combination of the variables, so it doesn't help with non-linearity. Adding the penalty actually increases bias, so it doesn't lower it.

It does reduce variance by shrinking the coefficients toward zero; indeed, that's why we introduced it.

It also does not incorporate interactions in any way, but I'm not going to deduct points for that answer since it turns out not to have been a familiar term.

Q4

2 Points

Consider the following two estimation procedures

$$\hat{\beta}_1 = \operatorname{argmin}_{\beta} \sum_{i=1}^n (y_i - x_i^T \beta)^2$$

$$\hat{\beta}_2 = \operatorname{argmin}_{\beta} \log \left(\sum_{i=1}^n (y_i - x_i^T \beta)^2 \right)$$

If both are evaluated on the same data X and Y , how will the estimates $\hat{\beta}_1$ and $\hat{\beta}_2$ compare?

- ☐ $\|\hat{\beta}_1\|_2^2 < \|\hat{\beta}_2\|_2^2$
- ☒ $\|\hat{\beta}_1\|_2^2 = \|\hat{\beta}_2\|_2^2$
- ☐ $\|\hat{\beta}_1\|_2^2 > \|\hat{\beta}_2\|_2^2$

Since one objective is just a monotone transformation of the other, the minima are achieved at the same values of β , so the resulting coefficient vectors are the same.

Q5

2 Points

Now consider the following two estimation procedures

$$\hat{\beta}_3 = \operatorname{argmin}_{\beta} \sum_{i=1}^n (y_i - x_i^T \beta)^2 + \|\beta\|_2^2$$

$$\hat{\beta}_4 = \operatorname{argmin}_{\beta} \log \left(\sum_{i=1}^n (y_i - x_i^T \beta)^2 \right) + \|\beta\|_2^2$$

If both are evaluated on the same data X and Y , how will the estimates $\hat{\beta}_3$ and $\hat{\beta}_4$ compare?

- ☐ $\|\hat{\beta}_3\|_2^2 < \|\hat{\beta}_4\|_2^2$
- ☐ $\|\hat{\beta}_3\|_2^2 = \|\hat{\beta}_4\|_2^2$
- ☒ $\|\hat{\beta}_3\|_2^2 > \|\hat{\beta}_4\|_2^2$

When we take the log of the sum of the squared residuals, we decrease the importance of that term relative to the ridge penalty. (This is a similar effect to increasing λ in usual ridge regression). The impact is that the optimization then cares more about the second term in the objective, and leads to a smaller coefficient vector.